07/10/26

Signal Processing

- Recap: Inverse z transform – inspection, partial fraction, series expansion
- Revisit pole-zero examples ✓
- z-transform properties: ROC needs attention
  - Linearity
  - ✓ Time shift. Ex: $X(z) = \frac{1}{z - \frac{1}{4}}$, $|z| > 1/4$
  - ✓ Multiplication and connection to DTFT
  - Conjugation
  - Time-reversal. Ex: $x[n] = a^{-n} u[-n]$
  - ✓ Convolution: $x_1[n] = a^n u[n]$, $x_2[n] = u[n]$. Assume $|a| < 1$.

- Recap: $X(z) = \frac{1}{1 - az^{-1}}$ $\quad |z| < |a|$

$$= \frac{z}{z - a} = \frac{z}{-a + z}$$

$$(-a + z) \;\Big|\; z \quad \Big|\; -a^{-1}z^{1} - a^{-2}z^{2} - a^{-3}z^{3} - \cdots$$

$$-z + a^{-1}z^{2}$$

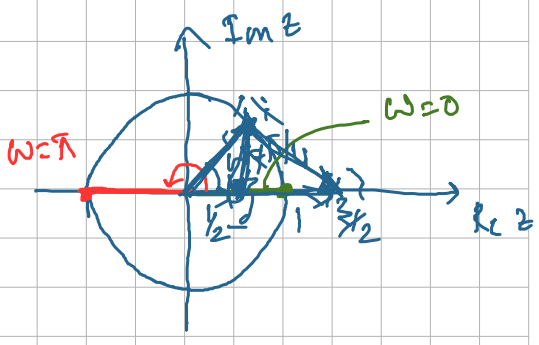
$$a^{-1}z^{2}$$

$$a^{-1}z^{2} - a^{-2}z^{3}$$

$$= -(a)^n \cdot z^{-n} \sum_{n=-\infty}^{-1} -(a)^n \cdot z^{-n}$$

$$\Rightarrow x[n] = -(a)^n u[-n-1]$$ ✓✓

- $H(z) = \frac{z - \frac{1}{2}}{z - \frac{3}{2}}$; $|z| < \frac{3}{2}$



- $|H(e^{j\omega})|_{\omega=0} = -1$
- $|H(e^{j\omega})|_{\omega=\pi} = 0.6$

- $\omega = \frac{\pi}{4} \Rightarrow \quad z = \frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}$

$$\Rightarrow \left| H(e^{j\omega}) \right|_{\omega=\frac{\pi}{4}} = \left| \frac{\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}} - \frac{1}{2}}{\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}} - \frac{3}{2}} \right| = 0.5 \quad \text{check}$$

- Experiment with the location of poles and zeros to give us the desired frequency response.

  Ex 1: If we want amplification at $\omega = 0$, push the pole to $z = 1.25$ or push the zero to $z = 0.25$

  Ex 2: If we want amplification at $\omega = \pi$, push the pole to $z = -0.75$

- Find $h[n]$ for the given $H(z)$; $|z| < \frac{3}{2}$

$$H(z) = \frac{z - \frac{1}{2}}{z - \frac{3}{2}} = z\,\frac{\left(1 - \frac{1}{2}z^{-1}\right)}{z\left(1 - \frac{3}{2}z^{-1}\right)} \Rightarrow$$

$$\Rightarrow h[n] = \frac{1}{3}\delta[n] - \left(\frac{3}{2}\right)^{n-1} \cdot u[-n-1]$$

$$H(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{3}{2}z^{-1}} = \frac{Y(z)}{X(z)}$$

$$\Rightarrow \left(1 - \tfrac{1}{2}z^{-1}\right)X(z) = \left(1 - \tfrac{3}{2}z^{-1}\right)Y(z)$$

$$x[n] - \frac{1}{2}x[n-1] = y[n] - \frac{3}{2}y[n-1]$$

$$y[n] = x[n] - \frac{1}{2}x[n-1] + \frac{3}{2}y[n-1]$$

- z-transform properties: Pay attention to ROC.

  - Time shift: If $x[n] \longleftrightarrow X(z)$, R_x

    $x[n-n_0] \longleftrightarrow ?$

$$\Rightarrow \mathcal{Z}\{x[n-n_0]\} = \sum_{n=-\infty}^{\infty} x[n-n_0] \cdot z^{-n}$$

$m = n - n_0$

$$= \sum_{m=-\infty}^{\infty} x[m] \, z^{-(m+n_0)}$$

$$= z^{-n_0} \sum_{m=-\infty}^{\infty} x[m] z^{-m}$$

$$= z^{-n_0} \cdot X(z) \quad ; \quad R_x \text{ with potential impact on any zeros or poles at } z=0,\ z=\infty.$$

- Find the inverse z-transform of

$$X(z) = \frac{1}{z - \frac{1}{4}}, \quad |z| > \frac{1}{4}$$

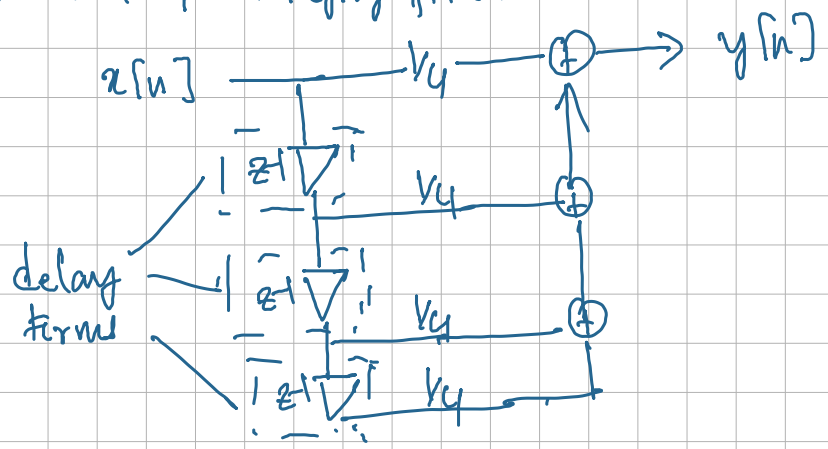
$$= \frac{z^{-1}}{1 - \frac{1}{4} z^{-1}}, \quad |z| > \frac{1}{4}$$

$$= z^{-1} \cdot \frac{1}{1 - \frac{1}{4} z^{-1}}, \quad |z| > \frac{1}{4}$$

$$\Rightarrow x[n] = x_0[n-1] \text{ where } x_0[n] = \left(\frac{1}{4}\right)^n u[n].$$

- A z^{-n_0} multiplication factor corresponds to a delay of n_0 units.

Recall the 4 tap averaging filter:



- "Modulation" $z_0^n x[n]$